

Solution Requirements

Date	16 September 2026
Team ID	SWTID-2026-3833
Project Name	AI StudyBuddy API
Maximum Marks	4 Marks

Step 1: Functional Requirements (FR)

S.No	Requirement Category	Requirement Description	Priority
1	Authentication	Users register and log in with name, email & password; passwords are hashed with bcryptjs and an access token + refresh token pair is issued via jsonwebtoken (JWT_ACCESS_SECRET).	High
2	Authorization levels	Role-Based Access Control via middleware/auth.js : a protect guard verifies the Bearer token on every private route, and an adminOnly guard restricts administrative endpoints to users with role: "admin".	High
3	External interfaces	Integrates with the Google Gemini API (@google/generative-ai , model gemini-2.5-flash) via a dedicated src/utls/gemini.js service for four flows: summary, flashcards, quiz, and study plan generation.	High
4	Transactions / data processing	Upload, retrieve and manage Material documents (title, raw content, original filename), each linked to its owning User ; each AI-generation call updates that same Material document with the generated result.	High
5	Reporting	Returns structured JSON for every material and AI-generation call, suitable for the React frontend (or any API client) to render.	Medium
6	Business rules	Each user's email must be unique; uploaded files are restricted to .txt , .md and .pdf with a 5 MB size limit (enforced in middleware/upload.js); every material and AI resource is scoped to the authenticated user who created it.	High
7	Compliance to laws or regulations	Passwords are never stored in plaintext (bcryptjs hashing) or returned in API responses; JWT secrets and the Gemini API key are kept in .env and excluded from version control via .gitignore .	High
8	Other	Centralized error handling (a global app.use((err, req, res, next) => ...) handler in index.js) returns consistent JSON error responses for auth failures, validation errors and unexpected errors.	Medium

Step 2: Non-Functional Requirements (NFR)

S.No	NFR Category	Requirement Description	Target Metric / Acceptance Criteria
1	Performance & Speed	Auth and Material CRUD endpoints (MongoDB Atlas-backed) should respond quickly; AI-generation endpoints depend on Gemini's external response latency.	Avg response < 800 ms for auth/material calls (see Performance Testing for measured samples)
2	Scalability	Stateless JWT authentication (access + refresh tokens) allows multiple Node.js instances to run behind a load balancer with no shared session state; MongoDB Atlas provides managed horizontal scaling for the database tier.	Designed to support growth beyond a single-instance deployment
3	Security & Data Privacy	bcryptjs password hashing, JWT-based access/refresh authentication, RBAC via adminOnly , and Mongoose schema validation (required fields) on every write.	No plaintext password storage; all inputs validated
4	Reliability & Availability	Centralized error-handling middleware plus a connectDB() guard in src/utls/db.js that only starts the Express listener after a confirmed MongoDB Atlas connection.	Server does not start serving requests without a live database connection
5	Usability & Accessibility	Consistent, well-documented JSON REST responses, verified with a shared Postman/Thunder Client collection covering auth, material and AI-generation endpoints.	Every endpoint documented and testable from the shared collection
6	Other (Maintainability)	Modular structure (controllers/ middleware/ models/ routes/ utls/) under src/ , so a new AI feature can be added without touching existing modules.	New feature addable as a new controller method + route + prompt template